Experiment - 11

Develope customized ping command to test server connectivity

Aim:

To develope a customized ping command using python to test the connectivity and response time

Procedure:

1) Import socket, struct, os, time and select modules

2) create a function to calculate ICMP checksum

3) Build ICMP echo request

4) create raw socket with socket IPPort

5) send the packet to the target hosting using sendto()

6) use select() to wait for a reply within a timeout period

7) recive the reply and unpack the ICMP header

commands / code

```
import socket, struct, time, os, select

def checksum(data)
s=0
for i in range (0, length(d), 2)
    s+= (d[i]<<8)+ d[i+1]
s = (s>>16)+ (s & 0xffff)
return ~s & 0xffff

def ping(host, c=4, t=2):
  pid= os.getpid() & 0xFFF
  r=[]
  try:
    dest= socket.gethostbyname(host)
    sock = socket.socket(socket.AF_INET)
for seq in range(c):
    h= struct.pack('bbHHh', 8, 0, 0, pid, seq)
    payload = b'ping packet'
    cs = checksum(header + payload)
    h= time.time()
    if r:
        rtt= (time.time() - t0)* 1000
        result.append(rtt)
```

8) calculate and display the round trip

9) send multiple packets and compute min max and averge RTT

10) display packet loss %

11) test with different host and record result

R 13 x 9 10

Result:

A customized ping utility was build successfully, ICMP echo request was sent, replies and RTT statistic include min, max & avg time displayed